

NETWORKING LAB REPORT

IPv6 Addressing & Basic Device Configuration

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Course: CCNA: Introduction to Networks

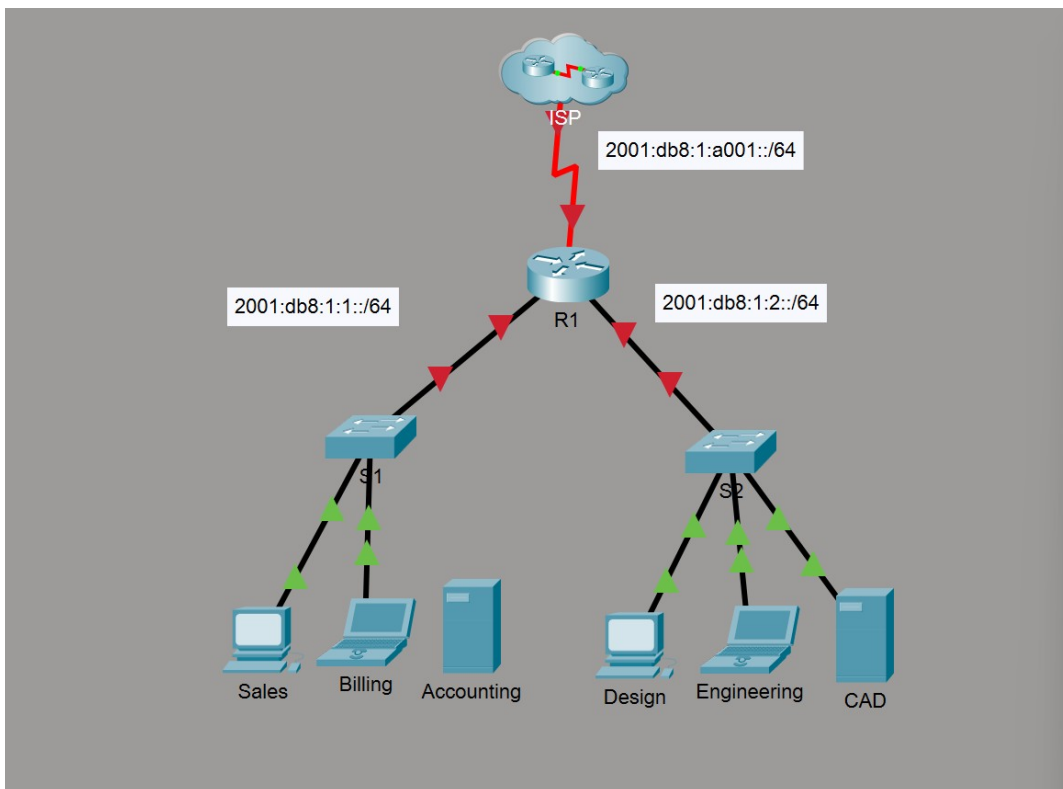
Group: IIoT-2401

Labs: 12.6.6 -Configure IPv6 Addressing | 12.4.5 -Basic Device Configuration

Lab 12.6.6 -Configure IPv6 Addressing

Objectives

The purpose of this lab is to practice configuring IPv6 addresses on a router, servers, and client PCs, and to verify end-to-end IPv6 connectivity across multiple network segments connected through router R1.



Addressing Table

Device	Interface	IPv6 Address / Prefix	Default Gateway
R1	G0/0	2001:DB8:1:1::1/64	N/A
R1	G0/0	fe80::1 (link-local)	N/A

R1	G0/1	2001:DB8:1:2::1/64	N/A
R1	G0/1	fe80::1 (link-local)	N/A
R1	S0/0/0	2001:DB8:1:A001::2/64	N/A
R1	S0/0/0	fe80::1 (link-local)	N/A
Sales	NIC	2001:DB8:1:1::2/64	fe80::1
Billing	NIC	2001:DB8:1:1::3/64	fe80::1
Accounting	NIC	2001:DB8:1:1::4/64	fe80::1
Design	NIC	2001:DB8:1:2::2/64	fe80::1
Engineering	NIC	2001:DB8:1:2::3/64	fe80::1
CAD	NIC	2001:DB8:1:2::4/64	fe80::1
ISP	S0/0/0	2001:DB8:1:A001::1/64	N/A

Part 1: Configure IPv6 Addressing on the Router

Q: What command enables the router to forward IPv6 packets? Why is it needed?

A: `ipv6 unicast-routing`. Unlike IPv4 where routing is enabled by default, in IPv6 the router will not forward packets between interfaces until this command is explicitly entered in global configuration mode.

Q: What is the difference between a global unicast address and a link-local address?

A: A **global unicast address** (e.g., 2001:DB8:1:1::1) is routable across the entire network and internet -it uniquely identifies a device globally. A **link-local address** (e.g., fe80::1) only works on the directly connected link (cable segment) and cannot be routed. It is used for neighbor discovery and as a default gateway for hosts.

Q: Why does each interface have TWO IPv6 addresses?

A: In IPv6, every interface needs at least two addresses: a global unicast address for network-wide communication, and a link-local address for local operations like neighbor discovery and serving as the default gateway for connected hosts. This is different from IPv4 where each interface typically has only one address.

Q: If you enter a wrong IPv6 address, can you just type the correct one over it?

A: No. In IPv6, entering a new address does not replace the old one -**both addresses will remain** on the interface. You must first remove the incorrect address using `no ipv6 address <wrong-address>`, then enter the correct one.

Q: What command verifies IPv6 addressing on the router?

A: `show ipv6 interface brief` -this displays all interfaces with their IPv6 addresses and up/down status in a concise table format.

R1 Configuration Commands

```
enable
configure terminal
ipv6 unicast-routing
```

```
interface GigabitEthernet 0/0
ipv6 address 2001:db8:1:1::1/64
ipv6 address fe80::1 link-local
no shutdown
exit

interface GigabitEthernet 0/1
ipv6 address 2001:db8:1:2::1/64
ipv6 address fe80::1 link-local
no shutdown
exit

interface Serial 0/0/0
ipv6 address 2001:db8:1:a001::2/64
ipv6 address fe80::1 link-local
no shutdown
exit

end
show ipv6 interface brief
copy running-config startup-config
```

```
R1
CLI

compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

R1>enable
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ipv6 unicast-routing
R1(config)#interface GigabitEthernet 0/0
R1(config-if)#ipv6 address 2001:db8:1:1::1/64
R1(config-if)#ipv6 address fe80::1 link-local
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to
up
exit
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1>enable
R1#interface GigabitEthernet 0/1
^
% Invalid input detected at '^' marker.

R1#
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface GigabitEthernet 0/1
R1(config-if)#ipv6 address 2001:db8:1:2::1/64
R1(config-if)#ipv6 address fe80::1 link-local
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to
up
exit
R1(config)#exit
R1(config)#interface Serial 0/0/0
R1(config-if)#ipv6 address 2001:db8:1:a001::2/64
R1(config-if)#ipv6 address fe80::1 link-local
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

R1(config-if)#exit
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
copy run start
Destination filename [startup-config]?

Copy Paste
```

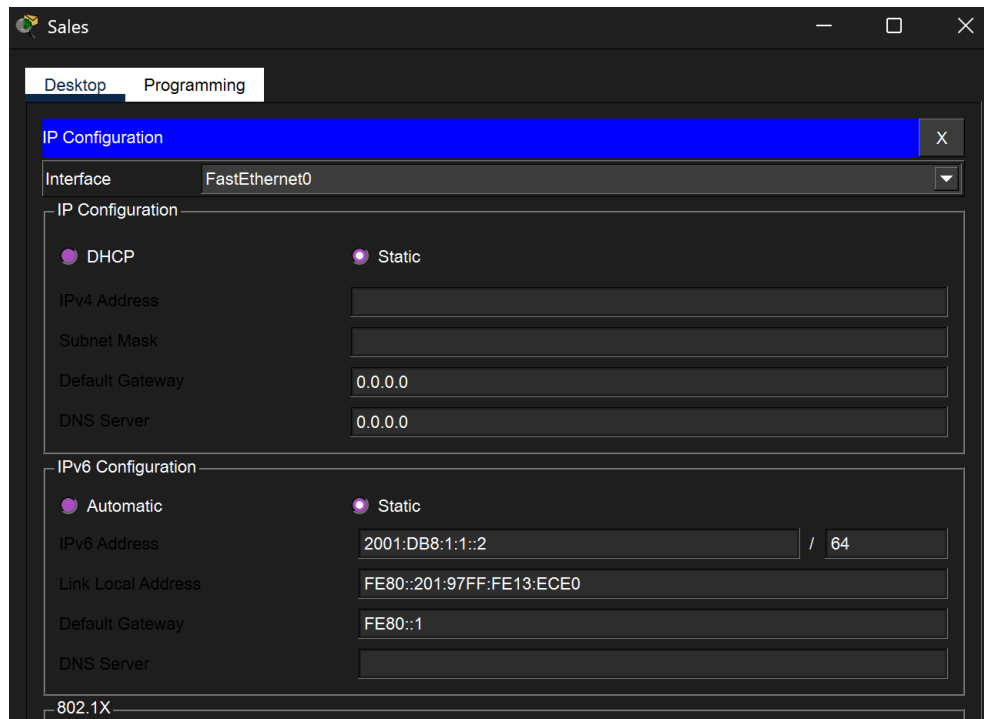
Top

Part 2 & 3: Configure IPv6 on Servers and Clients

Each device was configured via Desktop → IP Configuration with IPv6 set to **Static**. The IPv6 address, /64 prefix, and fe80::1 gateway were entered manually on all six devices.

Device	IPv6 Address	Prefix	Gateway
Sales	2001:DB8:1:1::2	/64	fe80::1
Billing	2001:DB8:1:1::3	/64	fe80::1
Accounting	2001:DB8:1:1::4	/64	fe80::1
Design	2001:DB8:1:2::2	/64	fe80::1

Engineering	2001:DB8:1:2::3	/64	fe80::1
CAD	2001:DB8:1:2::4	/64	fe80::1



Troubleshooting: Accounting Server

During testing, pings to Accounting (2001:DB8:1:1::4) consistently returned **Request timed out**. Upon investigation, it was discovered that the Accounting server was **not physically connected** to switch S1 -the Ethernet cable was missing from the lab file. This is a known issue with the Packet Tracer .pka file, not a configuration error. All other devices (Sales, Billing, Design, Engineering, CAD) were connected and functional.

Part 4: Test and Verify Network Connectivity

Ping Tests

Pings were executed from multiple clients to verify connectivity across both LAN segments and the WAN link to ISP:

From Sales: Ping to R1 (2001:db8:1:1::1) -**Success** (4/4, 0% loss). Ping to Accounting (::4) -**Failed** (missing cable).

From Billing: Ping to ISP (2001:db8:1:a001::1) -**Success**. Ping to Billing (::3) -**Success**. Ping to CAD (2001:db8:1:2::4) -**Success**.

From Engineering: Ping to Billing (::3) -**Success**. Ping to CAD (::4) -**Success**. Ping to ISP -**Success**.

```
Sales

Desktop Programming

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:db8:1:1::4

Pinging 2001:db8:1:1::4 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 2001:DB8:1:1::4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 2001:db8:1:1::1

Pinging 2001:db8:1:1::1 with 32 bytes of data:

Reply from 2001:DB8:1:1::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:1:1::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:1:1::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:1:1::1: bytes=32 time<1ms TTL=255

Ping statistics for 2001:DB8:1:1::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

Billing

Desktop Programming

Command Prompt

```
Pinging 2001:db8:1:1::4 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 2001:DB8:1:1::4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 2001:db8:1:a001::1

Pinging 2001:db8:1:a001::1 with 32 bytes of data:

Reply from 2001:DB8:1:A001::1: bytes=32 time=1ms TTL=254
Reply from 2001:DB8:1:A001::1: bytes=32 time=38ms TTL=254
Reply from 2001:DB8:1:A001::1: bytes=32 time=1ms TTL=254
Reply from 2001:DB8:1:A001::1: bytes=32 time=1ms TTL=254

Ping statistics for 2001:DB8:1:A001::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 38ms, Average = 10ms

C:\>ping 2001:db8:1:1::3

Pinging 2001:db8:1:1::3 with 32 bytes of data:

Reply from 2001:DB8:1:1::3: bytes=32 time=34ms TTL=128
Reply from 2001:DB8:1:1::3: bytes=32 time=14ms TTL=128
Reply from 2001:DB8:1:1::3: bytes=32 time=2ms TTL=128
Reply from 2001:DB8:1:1::3: bytes=32 time=12ms TTL=128

Ping statistics for 2001:DB8:1:1::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 34ms, Average = 15ms

C:\>ping 2001:db8:1:2::4

Pinging 2001:db8:1:2::4 with 32 bytes of data:

Reply from 2001:DB8:1:2::4: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:1:2::4: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:1:2::4: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:1:2::4: bytes=32 time<1ms TTL=127

Ping statistics for 2001:DB8:1:2::4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

Design

Desktop Programming

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:db8:1:1::3ping 2001:db8:1:2::4ping 2001:db8:
Invalid Command.

C:\>ping 2001:db8:1:1::3

Pinging 2001:db8:1:1::3 with 32 bytes of data:

Reply from 2001:DB8:1:1::3: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:1:1::3: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:1:1::3: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:1:1::3: bytes=32 time<1ms TTL=127

Ping statistics for 2001:DB8:1:1::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 2001:db8:1:2::4

Pinging 2001:db8:1:2::4 with 32 bytes of data:

Reply from 2001:DB8:1:2::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1:2::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1:2::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1:2::4: bytes=32 time<1ms TTL=128

Ping statistics for 2001:DB8:1:2::4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 2001:db8:1:a001::1

Pinging 2001:db8:1:a001::1 with 32 bytes of data:

Reply from 2001:DB8:1:A001::1: bytes=32 time=22ms TTL=254
Reply from 2001:DB8:1:A001::1: bytes=32 time=1ms TTL=254
Reply from 2001:DB8:1:A001::1: bytes=32 time=1ms TTL=254
Reply from 2001:DB8:1:A001::1: bytes=32 time=1ms TTL=254

Ping statistics for 2001:DB8:1:A001::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 22ms, Average = 6ms

C:\>
```

Web Browser Tests

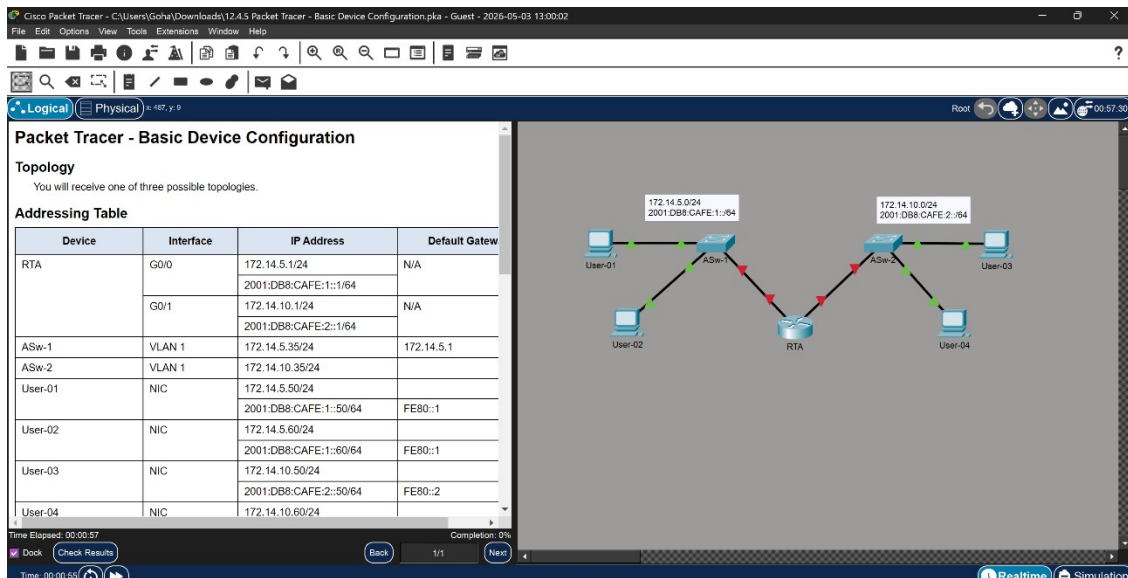
The CAD Server web page was successfully accessed from multiple clients by entering `2001:db8:1:2::4` in the URL bar. The page displayed "Welcome to the CAD server!" confirming HTTP connectivity over IPv6. The Accounting web page could not be accessed due to the missing cable issue.

The screenshot displays a Cisco Packet Tracer simulation environment. At the top, a web browser window is open, showing the "CAD Server" page with the message "Welcome to the CAD server!". The browser's address bar contains the URL `http://2001:db8:1:2::4`, and the "Go" button is highlighted. Below the browser, the network topology is visible. A central router, R1, is connected to an ISP cloud (2001:db8:1:a001::64) and two switches. The left switch is connected to three desktops labeled "Sales", "Billing", and "Accounting". The right switch is connected to three desktops labeled "Design", "Engineering", and "CAD". The IP addresses for the switches are 2001:db8:1:1::64 and 2001:db8:1:2::64. A "PT Activity" window is open on the right, showing instructions for testing network connectivity, including steps for opening server web pages and pinging the ISP. The bottom status bar indicates the simulation is running in "Realtime" mode.

Lab 12.4.5 -Basic Device Configuration

Objectives

This lab demonstrates configuring a dual-stack (IPv4 + IPv6) network with a router, switch, and four PCs. Tasks include setting hostnames, security passwords, MOTD banners, interface addressing with descriptions, and verifying full end-to-end connectivity over both protocols.



Complete Addressing Table

Device	Interface	IP Address	Default Gateway
RTA	G0/0	172.14.5.1/24	N/A
RTA	G0/0	2001:DB8:CAFE:1::1/64	N/A
RTA	G0/1	172.14.10.1/24	N/A
RTA	G0/1	2001:DB8:CAFE:2::1/64	N/A
ASw-1	VLAN 1	172.14.5.35/24	172.14.5.1
ASw-2	VLAN 1	172.14.10.35/24	172.14.10.1
User-01	NIC (IPv4)	172.14.5.50/24	172.14.5.1
User-01	NIC (IPv6)	2001:DB8:CAFE:1::50/64	fe80::1
User-02	NIC (IPv4)	172.14.5.60/24	172.14.5.1
User-02	NIC (IPv6)	2001:DB8:CAFE:1::60/64	fe80::1
User-03	NIC (IPv4)	172.14.10.50/24	172.14.10.1
User-03	NIC (IPv6)	2001:DB8:CAFE:2::50/64	fe80::2
User-04	NIC (IPv4)	172.14.10.60/24	172.14.10.1
User-04	NIC (IPv6)	2001:DB8:CAFE:2::60/64	fe80::2

Q: Why is the IPv6 gateway fe80::1 for User-01/02 but fe80::2 for User-03/04?

A: Because User-01 and User-02 are connected to RTA's G0/0 interface, which has link-local address fe80::1. User-03 and User-04 are connected to G0/1, which has link-local address fe80::2. Each interface has its own link-local address, and hosts use the link-local address of the router interface on their local segment as their gateway.

RTA Router Configuration

```
enable
configure terminal
hostname RTA
banner motd #Unauthorized access is prohibited.#
enable secret class
line console 0
password cisco
login
exit
line vty 0 4
password cisco
login
exit
service password-encryption
ipv6 unicast-routing

interface GigabitEthernet 0/0
ip address 172.14.5.1 255.255.255.0
ipv6 address 2001:DB8:CAFE:1::1/64
ipv6 address fe80::1 link-local
description LAN connection to ASw-1
no shutdown
exit

interface GigabitEthernet 0/1
ip address 172.14.10.1 255.255.255.0
ipv6 address 2001:DB8:CAFE:2::1/64
ipv6 address fe80::2 link-local
description LAN connection to ASw-2
no shutdown
exit

end
copy running-config startup-config
```

```
RTA
Physical  CLI

Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname RTA
RTA(config)#banner motd #Unauthorized access is prohibited.#
RTA(config)#enable secret class
RTA(config)#line console 0
RTA(config-line)#password cisco
RTA(config-line)#login
RTA(config-line)#exit
RTA(config)#line vty 0 4
RTA(config-line)#password cisco
RTA(config-line)#login
RTA(config-line)#exit
RTA(config)#service password-encryption
RTA(config)#ipv6 unicast-routing
RTA(config)#interface GigabitEthernet 0/0
RTA(config-if)#ip address 172.14.5.1 255.255.255.0
RTA(config-if)#
RTA(config-if)#ipv6 address 2001:DB8:CAFE:1::1/64
RTA(config-if)#ipv6 address fe80::1 link-local
RTA(config-if)#description LAN connection to ASw-1
RTA(config-if)#no shutdown

RTA(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up
RTA(config-if)#exit
RTA(config)#interface GigabitEthernet 0/1
RTA(config-if)#ip address 172.14.10.1 255.255.255.0
RTA(config-if)#ipv6 address 2001:DB8:CAFE:2::1/64
RTA(config-if)#ipv6 address fe80::2 link-local
RTA(config-if)#description LAN connection to ASw-2
RTA(config-if)#no shutdown

RTA(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1,
changed state to up
RTA(config-if)#exit
RTA(config)#end
RTA#
%SYS-5-CONFIG_I: Configured from console by console

RTA#copy running-config startup-config
Destination filename [startup-config]?
```

ASw-2 Switch Configuration

```
enable
configure terminal
hostname ASw-2
banner motd #Unauthorized access is prohibited.#
enable secret class
line console 0
password cisco
login
exit
line vty 0 15
password cisco
login
exit
service password-encryption
interface vlan 1
ip address 172.14.10.35 255.255.255.0
description Management interface
no shutdown
exit
ip default-gateway 172.14.10.1
end
copy running-config startup-config
```

```
state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/24, changed state to up

Switch>Enable
Switch#conf t
Enter configuration commands, one per line.  End with
CNTL/Z.
Switch(config)#hostname ASw-2
ASw-2(config)#banner motdbanner motd #Unauthorized access
is prohibited.#enable secret class
^
% Invalid input detected at '^' marker.

ASw-2(config)#banner motd #Unauthorized access is
prohibited.#
ASw-2(config)#enable secret class
ASw-2(config)#line console 0
ASw-2(config-line)#password cisco
ASw-2(config-line)#login
ASw-2(config-line)#exit
ASw-2(config)#line vty 0 15
ASw-2(config-line)#password cisco
ASw-2(config-line)#login
ASw-2(config-line)#exit
ASw-2(config)#service password-encryption
ASw-2(config)#interface vlan 1
ASw-2(config-if)#ip address 172.14.10.35 255.255.255.0
ASw-2(config-if)#description Management interface
ASw-2(config-if)#no shutdown

ASw-2(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1,
changed state to up

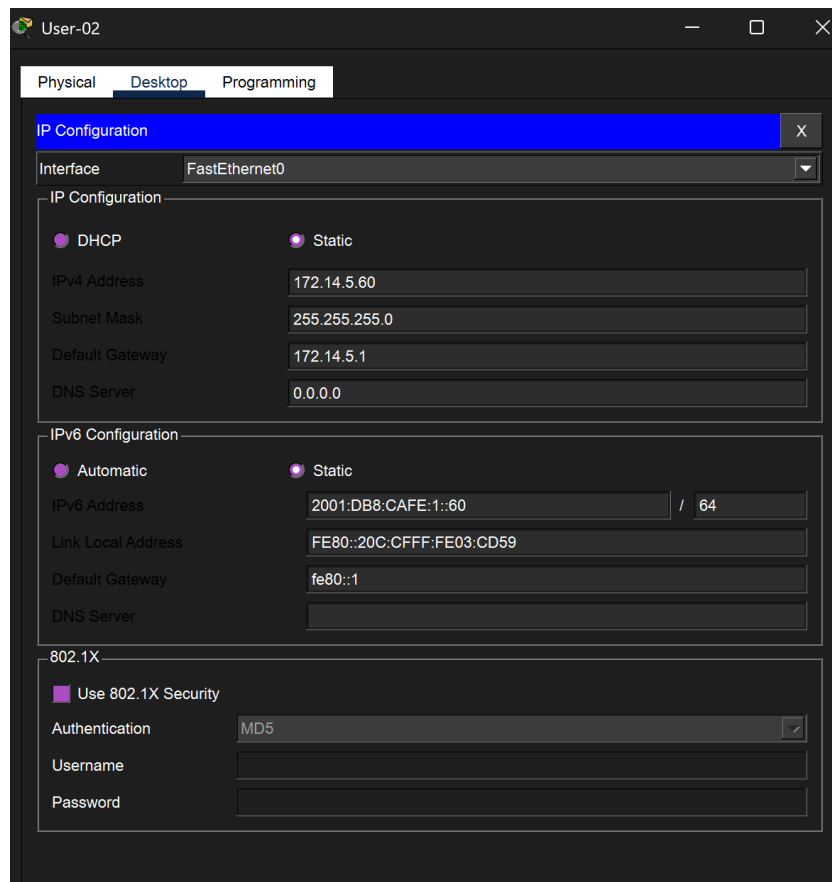
ASw-2(config-if)#exit
ASw-2(config)#ip default-gateway 172.14.10.1
ASw-2(config)#end
ASw-2#
%SYS-5-CONFIG_I: Configured from console by console

ASw-2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
ASw-2#
```

Copy Paste

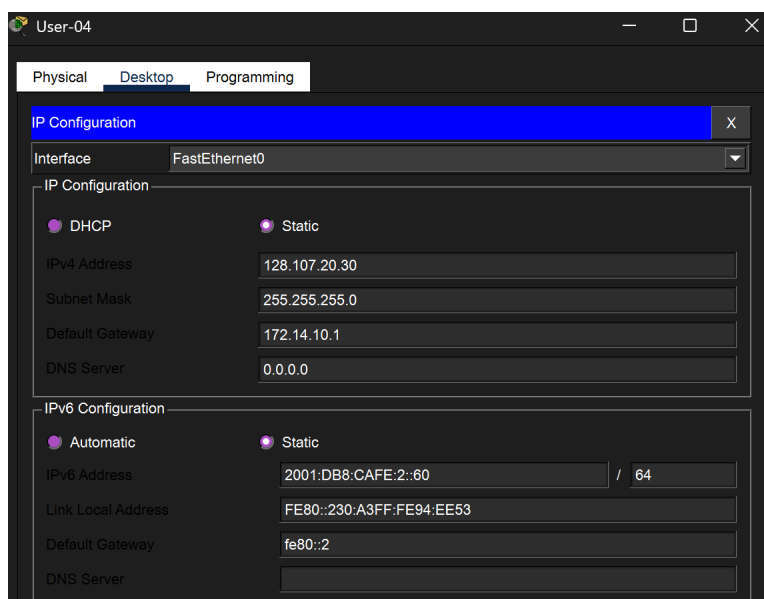
Host Configuration

All four users were configured via Desktop → IP Configuration with both IPv4 (Static) and IPv6 (Static) addresses.



Troubleshooting: User-04 Incorrect IPv4 Address

During verification, User-04 was found to have an incorrect IPv4 address of 128.107.20.30 pre-configured in the lab file. This address belongs to a completely different network (128.107.x.x) and does not match the required 172.14.10.0/24 subnet. The address was corrected to 172.14.10.60 with subnet mask 255.255.255.0 and default gateway 172.14.10.1. After correction, IPv4 connectivity was restored.



Verification: Connectivity Tests

IPv4 Ping Tests (from User-01)

Ping to 172.14.5.1 (RTA G0/0, local gateway) -**Success** (4/4, 0% loss).

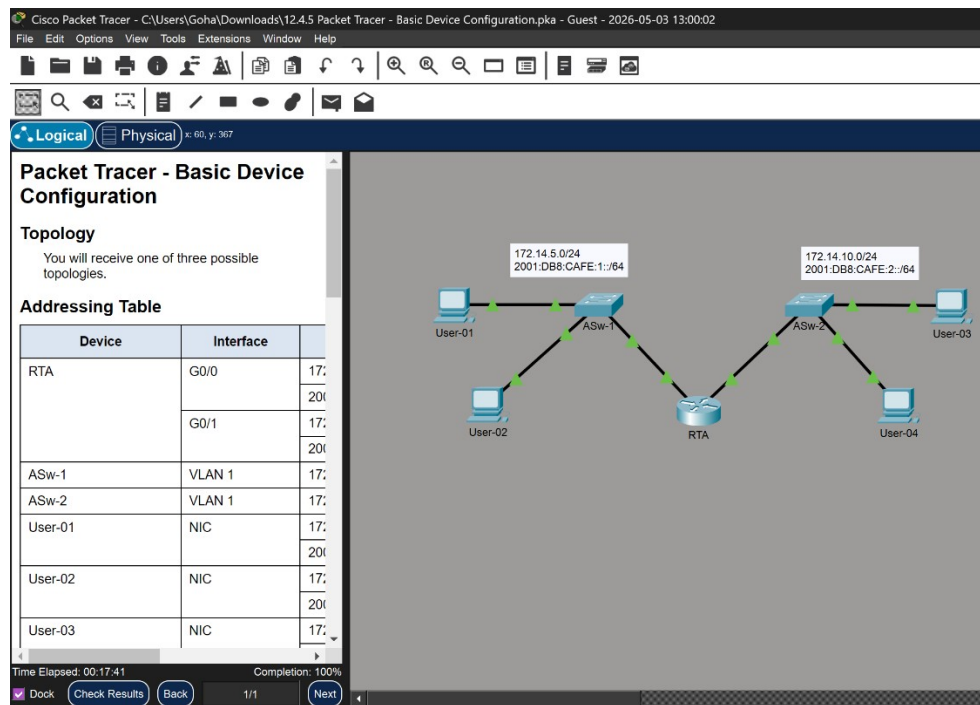
Ping to 172.14.10.50 (User-03, remote LAN) -**Success** (4/4 after initial ARP timeout).

IPv6 Ping Tests (from User-01)

Ping to 2001:DB8:CAFE:1::1 (RTA G0/0) -**Success** (4/4, 0% loss).

Ping to 2001:DB8:CAFE:2::50 (User-03, cross-LAN) -**Success**, confirming dual-stack routing works correctly.

User-01	User-01
<pre>Physical Desktop Programming Command Prompt C:\>ping 172.14.5.1 Pinging 172.14.5.1 with 32 bytes of data: Reply from 172.14.5.1: bytes=32 time=2ms TTL=255 Reply from 172.14.5.1: bytes=32 time<1ms TTL=255 Reply from 172.14.5.1: bytes=32 time<1ms TTL=255 Reply from 172.14.5.1: bytes=32 time<1ms TTL=255 Ping statistics for 172.14.5.1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 2ms, Average = 0ms C:\>ping 172.14.10.50 Pinging 172.14.10.50 with 32 bytes of data: Request timed out. Reply from 172.14.10.50: bytes=32 time<1ms TTL=127 Reply from 172.14.10.50: bytes=32 time<1ms TTL=127 Reply from 172.14.10.50: bytes=32 time<1ms TTL=127 Ping statistics for 172.14.10.50: Packets: Sent = 4, Received = 3, Lost = 1 (25% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms C:\>ping 172.14.10.50 Pinging 172.14.10.50 with 32 bytes of data: Reply from 172.14.10.50: bytes=32 time<1ms TTL=127 Reply from 172.14.10.50: bytes=32 time<1ms TTL=127 Reply from 172.14.10.50: bytes=32 time<1ms TTL=127 Reply from 172.14.10.50: bytes=32 time<1ms TTL=127 Ping statistics for 172.14.10.50: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms C:\>ping 2001:DB8:CAFE:1::1 Pinging 2001:DB8:CAFE:1::1 with 32 bytes of data: Reply from 2001:DB8:CAFE:1::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:CAFE:1::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:CAFE:1::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:CAFE:1::1: bytes=32 time<1ms TTL=255 Ping statistics for 2001:DB8:CAFE:1::1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms C:\>ping 2001:DB8:CAFE:2::50 Pinging 2001:DB8:CAFE:2::50 with 32 bytes of data: Reply from 2001:DB8:CAFE:2::50: bytes=32 time=10ms TTL=127 Reply from 2001:DB8:CAFE:2::50: bytes=32 time<1ms TTL=127 Reply from 2001:DB8:CAFE:2::50: bytes=32 time<1ms TTL=127 Reply from 2001:DB8:CAFE:2::50: bytes=32 time=16ms TTL=127 Ping statistics for 2001:DB8:CAFE:2::50: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 16ms, Average = 6ms C:\></pre>	<pre>Physical Desktop Programming Command Prompt Request timed out. Reply from 172.14.10.50: bytes=32 time<1ms TTL=127 Reply from 172.14.10.50: bytes=32 time<1ms TTL=127 Reply from 172.14.10.50: bytes=32 time<1ms TTL=127 Ping statistics for 172.14.10.50: Packets: Sent = 4, Received = 3, Lost = 1 (25% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms C:\>ping 172.14.10.50 Pinging 172.14.10.50 with 32 bytes of data: Reply from 172.14.10.50: bytes=32 time<1ms TTL=127 Reply from 172.14.10.50: bytes=32 time<1ms TTL=127 Reply from 172.14.10.50: bytes=32 time<1ms TTL=127 Reply from 172.14.10.50: bytes=32 time<1ms TTL=127 Ping statistics for 172.14.10.50: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms C:\>ping 2001:DB8:CAFE:1::1 Pinging 2001:DB8:CAFE:1::1 with 32 bytes of data: Reply from 2001:DB8:CAFE:1::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:CAFE:1::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:CAFE:1::1: bytes=32 time<1ms TTL=255 Reply from 2001:DB8:CAFE:1::1: bytes=32 time<1ms TTL=255 Ping statistics for 2001:DB8:CAFE:1::1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms C:\>ping 2001:DB8:CAFE:2::50 Pinging 2001:DB8:CAFE:2::50 with 32 bytes of data: Reply from 2001:DB8:CAFE:2::50: bytes=32 time=10ms TTL=127 Reply from 2001:DB8:CAFE:2::50: bytes=32 time<1ms TTL=127 Reply from 2001:DB8:CAFE:2::50: bytes=32 time<1ms TTL=127 Reply from 2001:DB8:CAFE:2::50: bytes=32 time=16ms TTL=127 Ping statistics for 2001:DB8:CAFE:2::50: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 16ms, Average = 6ms C:\></pre>



Conclusion

These two labs covered the fundamentals of IPv6 deployment in Cisco networks. In Lab 12.6.6, we configured a pure IPv6 environment: enabling `ipv6 unicast-routing` on the router, assigning both global unicast and link-local addresses to each interface, and configuring six end devices with static IPv6 addresses. Key takeaways include the requirement for two addresses per interface in IPv6, the use of link-local addresses as default gateways, and the need to explicitly remove incorrect IPv6 addresses rather than overwriting them. In Lab 12.4.5, we implemented a dual-stack configuration supporting both IPv4 and IPv6 simultaneously. This included full router security (passwords, banner, encryption), interface descriptions for documentation, and switch management addressing. Troubleshooting revealed a pre-configured error on User-04 (wrong IPv4 address from a different subnet), which was corrected to restore connectivity. Both labs achieved 100% completion, confirming proper addressing and full end-to-end reachability across all network segments.

—End of Report—